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Course Code: CBS3005

**Course Title: Cloud, Microservices and
Applications**

Class Number: VL2025260105679

Lab Assessment - 5

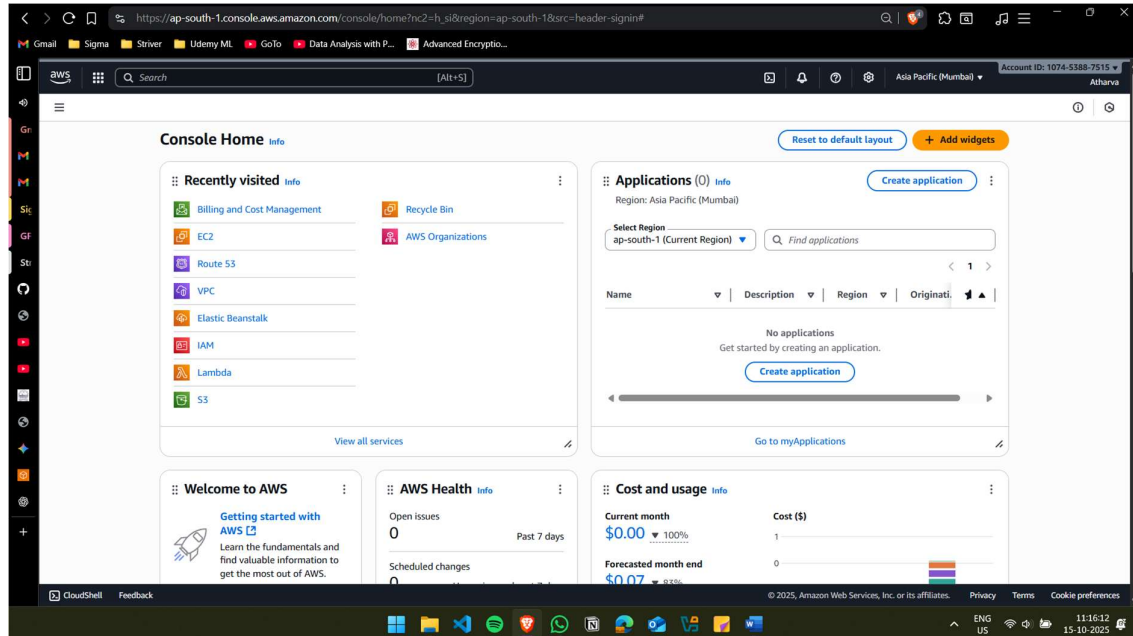
Name: Atharva Khachane

Reg No: 22BBS0155

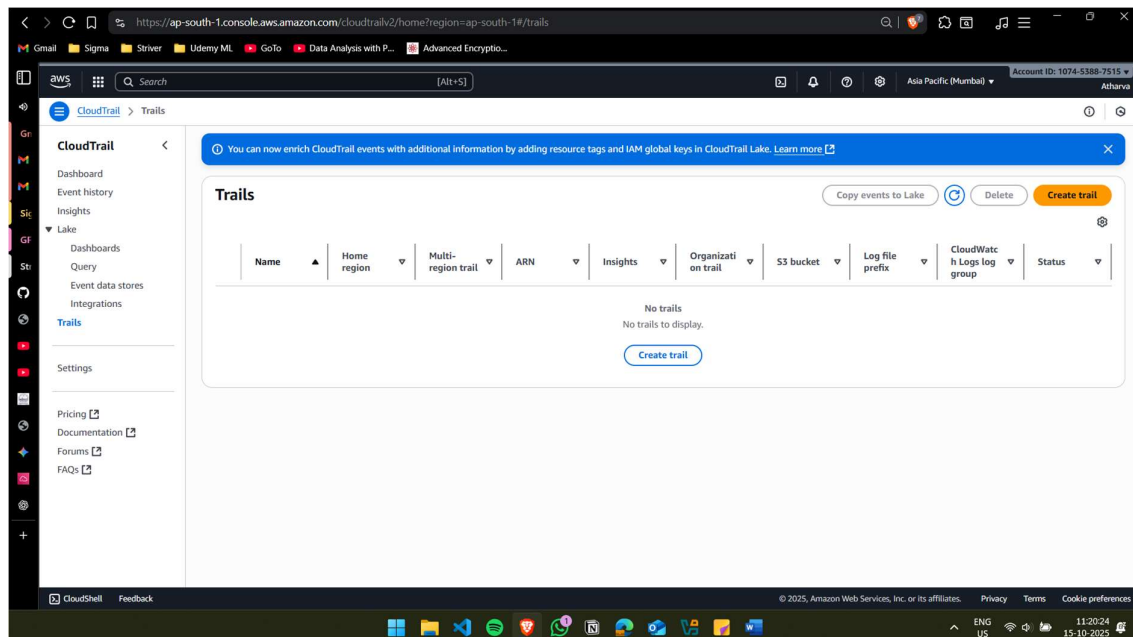
Q. Analyze API Activities using AWS CloudTrail

Description: Enable AWS CloudTrail to monitor and log all API activities in the AWS account. Also, analyze events to detect unauthorized or unusual access.

Initial:



Go to CloudTrail Console → Trails → Create Trail.



Choose:

Trail name: MySecurityTrail

Apply trail to All regions

Store logs in a new S3 bucket

Enable CloudWatch Logs Integration (optional).

The screenshot shows the 'Review and create' step of the AWS CloudTrail console. The trail is named 'MySecurityTrail' and is configured as a multi-region trail. The trail log location is set to 'aws-cloudtrail-logs-107453887515-a59c8eeb/AWSLogs/107453887515/'. Log file validation is enabled, and log file SSE-KMS encryption is not enabled. The trail is not applied to the organization. The CloudWatch Logs section shows that no log groups are configured. The Tags section is empty.

Review and create

Step 1: Choose trail attributes

General details

- Trail name: MySecurityTrail
- Multi-region trail: Yes
- Apply trail to my organization: Not enabled
- Trail log location: aws-cloudtrail-logs-107453887515-a59c8eeb/AWSLogs/107453887515/
- Log file validation: Enabled
- SNS notification delivery: Disabled
- Log file SSE-KMS encryption: Not enabled

CloudWatch Logs

No CloudWatch Logs log groups
CloudWatch Logs is not configured for this trail

Tags

Key	Value
No tags	

The screenshot shows the 'Trails' page in the AWS CloudTrail console. A green banner at the top indicates 'Trail successfully created'. A blue banner below it provides information about enriching events with resource tags and IAM global keys. The main table lists the trail 'MySecurityTrail' with its attributes.

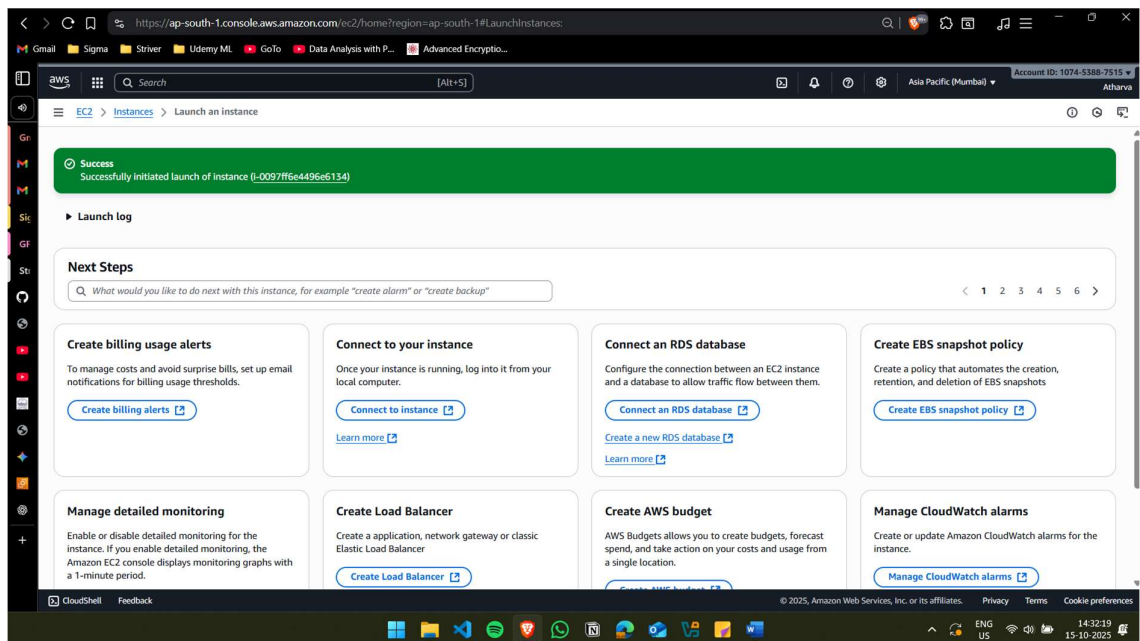
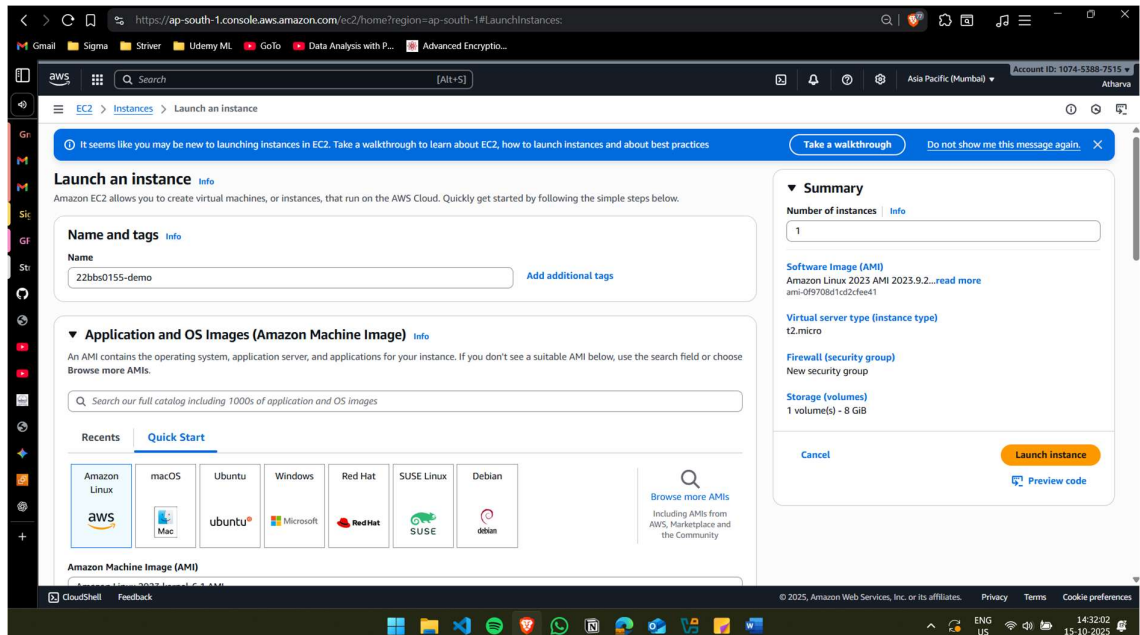
Trails

Copy events to Lake Delete Create trail

Name	Home region	Multi-region trail	ARN	Insights	Organization trail	S3 bucket	Log file prefix	CloudWatch Logs log group	Status
MySecurityTrail	Asia Pacific (Mumbai)	Yes	arn:aws:cloudtrail:ap-south-1:107453887515:trail/MySecurityTrail	Disabled	No	aws-cloudtrail-logs-107453887515-a59c8eeb	-	-	Logging

Perform a few activities in your AWS account:

Launch or stop an EC2 instance.



Stop the new instance

Successfully initiated termination (deletion) of i-0097ff6e4496e6134

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive) All states

Instance state = running Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 address
22bbs0155-demo	i-0097ff6e4496e6134	Running	t2.micro	Initializing	View alarms +	ap-south-1a	ec2-65-2-151-225.ap-s...	65

i-0097ff6e4496e6134 (22bbs0155-demo)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-0097ff6e4496e6134

IPv6 address -

Hostname type IP name: ip-172-31-35-216.ap-south-1.compute.internal

Public IPv4 address 65.2.151.225 [open address]

Instance state Running

Private IPv4 addresses 172.31.35.216

Public DNS ec2-65-2-151-225.ap-south-1.compute.amazonaws.com [open address]

Private IP DNS name (IPv4 only) ip-172-31-35-216.ap-south-1.compute.internal

Create an S3 bucket.

Successfully created bucket "22bbs0155-demo"
To upload files and folders, or to configure additional bucket settings, choose **View details**.

General purpose buckets (3) Info

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
22bbs0155-demo	Asia Pacific (Mumbai) ap-south-1	October 15, 2025, 14:33:15 (UTC+05:30)
aws-cloudtrail-logs-107453887515-a59c8eeb	Asia Pacific (Mumbai) ap-south-1	October 15, 2025, 14:30:42 (UTC+05:30)
elasticbeanstalk-ap-south-1-107453887515	Asia Pacific (Mumbai) ap-south-1	September 9, 2025, 23:13:20 (UTC+05:30)

Account snapshot Info

Updated daily
Storage Lens provides visibility into storage usage and activity trends.

External access summary - new Info

Updated daily
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

Return to CloudTrail → Event History and view recent events.

The screenshot shows the AWS CloudTrail console's Event History page. The left sidebar contains navigation links for CloudTrail, Dashboard, Insights, Lake, Dashboards, Query, Event data stores, Integrations, Trails, Settings, Pricing, Documentation, Forums, and FAQs. The main content area displays a table of events with columns for Event name, Event time, User name, Event source, Resource type, and Resource name. The table lists several events, including SharedSnapshotVolu..., RunInstances, AuthorizeSecurityGro..., CreateSecurityGroup, PutBucketEncryption, CreateTrail, StartLogging, PutEventSelectors, PutBucketPolicy, CreateBucket, and CreateServiceLinked... The bottom of the table indicates 0 / 5 events selected.

Event name	Event time	User name	Event source	Resource type	Resource name
SharedSnapshotVolu...	October 15, 2025, 14:32:17 (UT...	-	ec2.amazonaws.com	-	-
RunInstances	October 15, 2025, 14:32:15 (UT...	root	ec2.amazonaws.com	AWS::EC2::VPC, AWS::E...	vpc-010499e03ddb3d905, ami-0f9708d1...
AuthorizeSecurityGro...	October 15, 2025, 14:32:14 (UT...	root	ec2.amazonaws.com	AWS::EC2::SecurityGroup	sg-057da11f3a56c6aff
CreateSecurityGroup	October 15, 2025, 14:32:13 (UT...	root	ec2.amazonaws.com	AWS::EC2::VPC, AWS::E...	vpc-010499e03ddb3d905, launch-wizard...
PutBucketEncryption	October 15, 2025, 14:30:43 (UT...	root	s3.amazonaws.com	AWS::S3::Bucket	aws-cloudtrail-logs-107453887515-a59c...
CreateTrail	October 15, 2025, 14:30:43 (UT...	root	cloudtrail.amazonaws.c...	AWS::CloudTrail::Trail, ...	arn:aws:cloudtrail:ap-south-1:107453887...
StartLogging	October 15, 2025, 14:30:43 (UT...	root	cloudtrail.amazonaws.c...	AWS::CloudTrail::Trail	arn:aws:cloudtrail:ap-south-1:107453887...
PutEventSelectors	October 15, 2025, 14:30:42 (UT...	root	cloudtrail.amazonaws.c...	AWS::CloudTrail::Trail	arn:aws:cloudtrail:ap-south-1:107453887...
PutBucketPolicy	October 15, 2025, 14:30:42 (UT...	root	s3.amazonaws.com	AWS::S3::Bucket	aws-cloudtrail-logs-107453887515-a59c...
CreateBucket	October 15, 2025, 14:30:42 (UT...	root	s3.amazonaws.com	AWS::S3::Bucket	aws-cloudtrail-logs-107453887515-a59c...
CreateServiceLinked...	October 15, 2025, 11:17:57 (UT...	-	cloudtrail.amazonaws.c...	-	-

Identify:

Who performed the action (IAM user or role)

Time of action

Source IP address

Affected resource

Following is the example of createBucket event

The screenshot shows the AWS CloudTrail console's Event History page with the 'CreateBucket' event selected. The left sidebar is the same as the previous screenshot. The main content area displays the details of the event, including the Event time, User name, Event name, Event source, AWS access key, Source IP address, Event ID, Request ID, AWS region, Error code, and Read-only status. Below the details, there is a section for 'Resources referenced (1)' showing the resource type, resource name, and the AWS Config resource timeline. At the bottom, there is a section for 'Event record' showing the JSON view of the event.

Details

Event time	October 15, 2025, 14:33:14 (UTC+05:30)	AWS access key	ASIA85B6GQ45AYQLGQ	AWS region	ap-south-1
User name	root	Source IP address	182.66.218.121	Error code	-
Event name	CreateBucket	Event ID	c11bb5a3-c762-45b5-a666-1cc4e0c089eb	Read-only	false
Event source	s3.amazonaws.com	Request ID	HGKZ2817H4P7BKC6		

Resources referenced (1)

Resource type	Resource name	AWS Config resource timeline
AWS::S3::Bucket	22bbs0155-demo	Enable AWS Config resource recording

Event record

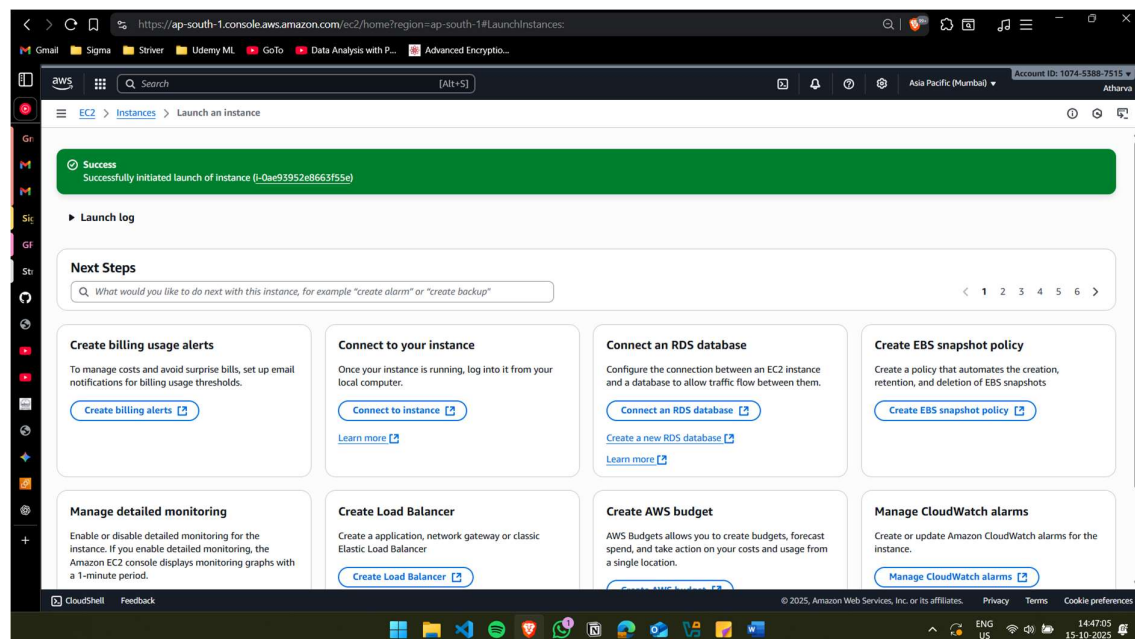
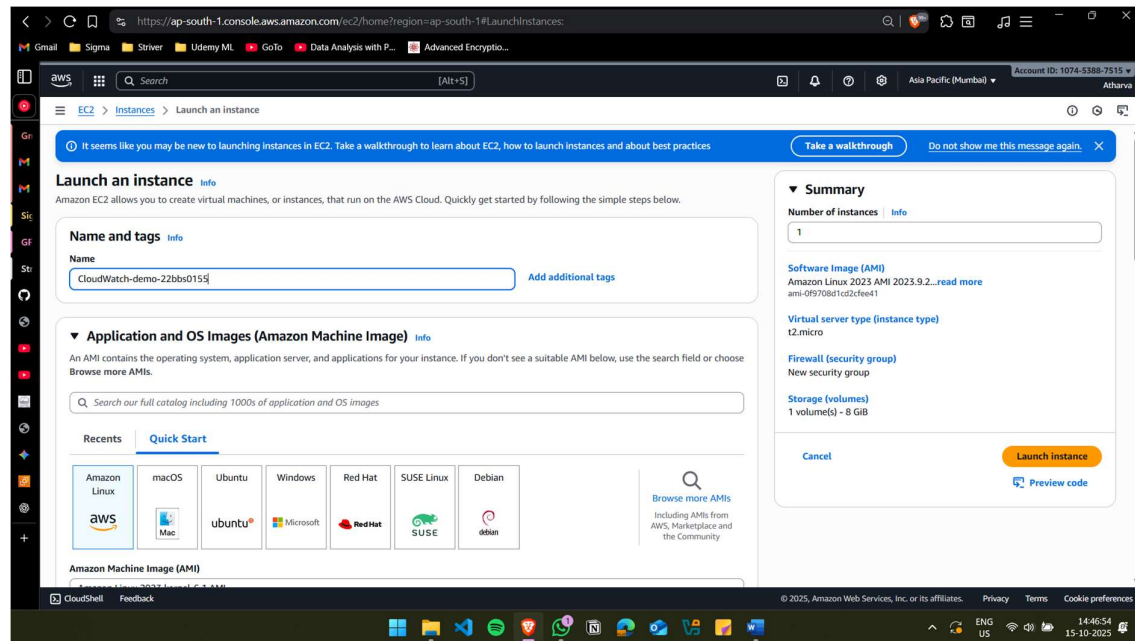
JSON view

```
1 {
2   "eventVersion": "1.11",
3   "userIdentity": {
4     "type": "Root"
```

Q. Configure CloudWatch for EC2 Instance Monitoring

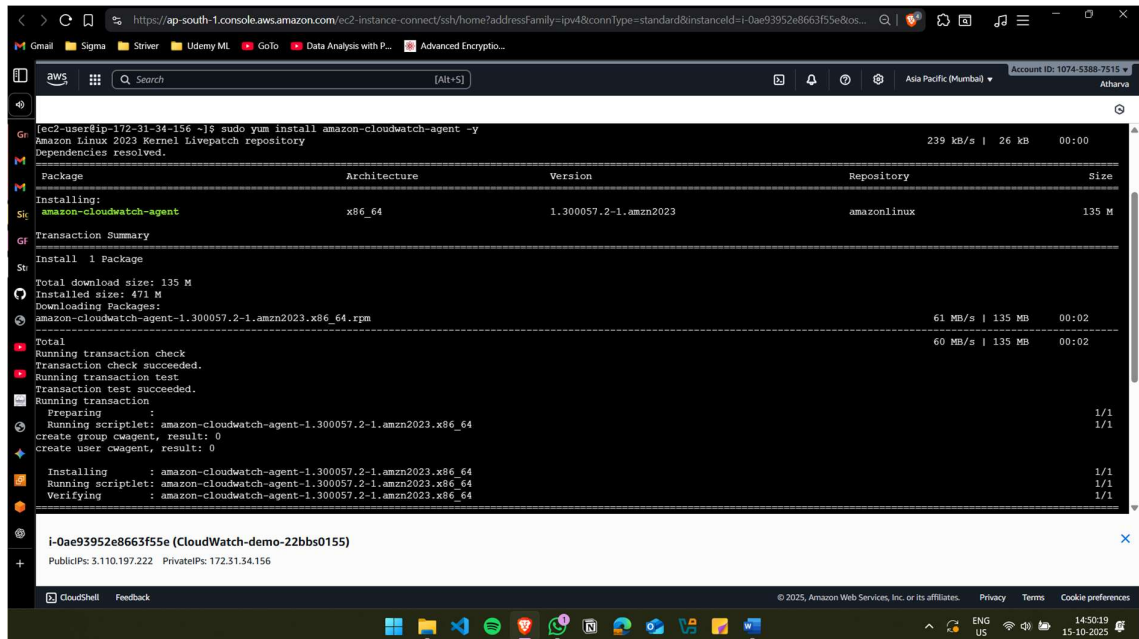
Description: Create and monitor a simple EC2 instance using Amazon CloudWatch , configure alarms, monitor system metrics, and analyze performance data to understand how CloudWatch helps in proactive threat and performance monitoring.

Launch a t2.micro EC2 instance (Free Tier eligible) running Amazon Linux 2.



Install and configure the CloudWatch Agent using:

```
sudo yum install amazon-cloudwatch-agent -y
```



```
[ec2-user@ip-172-31-34-156 ~]$ sudo yum install amazon-cloudwatch-agent -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
amazon-cloudwatch-agent x86_64 1.300057.2-1.amzn2023 amazonlinux 135 M
Transaction Summary
Install 1 Package
Total download size: 135 M
Installed size: 471 M
Downloading Packages:
amazon-cloudwatch-agent-1.300057.2-1.amzn2023.x86_64.rpm 61 MB/s | 135 MB 00:02
Total 60 MB/s | 135 MB 00:02
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction:
Preparing:
Running scriptlet: amazon-cloudwatch-agent-1.300057.2-1.amzn2023.x86_64
create group cwagent, result: 0
create user cwagent, result: 0
Installing : amazon-cloudwatch-agent-1.300057.2-1.amzn2023.x86_64 1/1
Running scriptlet: amazon-cloudwatch-agent-1.300057.2-1.amzn2023.x86_64 1/1
Verifying : amazon-cloudwatch-agent-1.300057.2-1.amzn2023.x86_64 1/1

i-Oae93952e8663f55e (CloudWatch-demo-22bbs0155)
PublicIPs: 3.110.197.222 PrivateIPs: 172.31.34.156
```

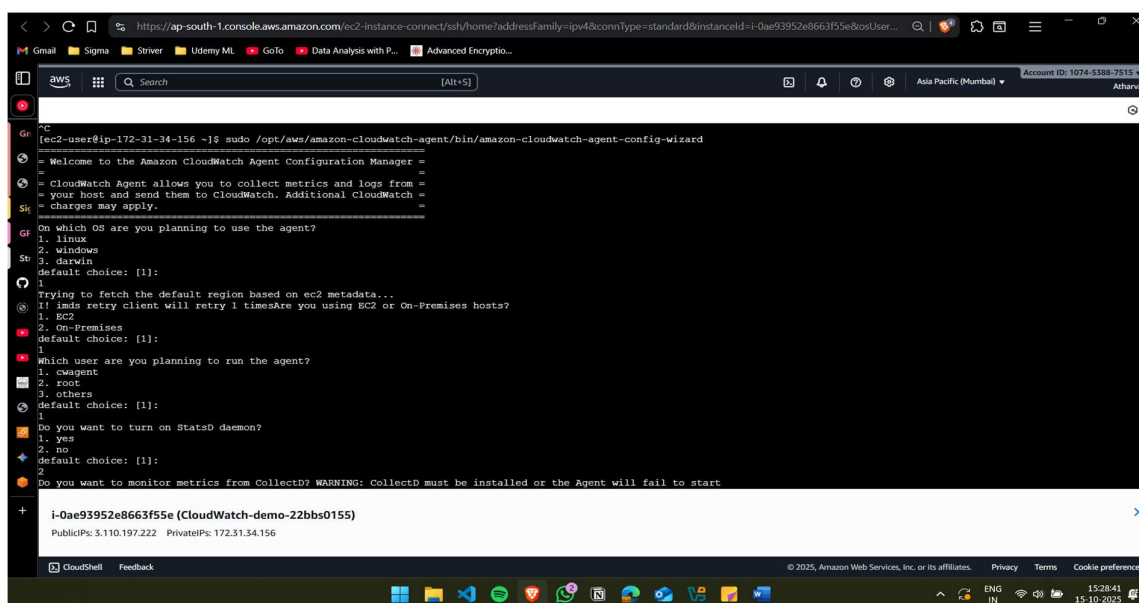
```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard
```

Choose to monitor:

CPU utilization

Memory usage

Disk I/O



```
[ec2-user@ip-172-31-34-156 ~]$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard
= Welcome to the Amazon CloudWatch Agent Configuration Manager =
= CloudWatch Agent allows you to collect metrics and logs from =
= your host and send them to CloudWatch. Additional CloudWatch =
= charges may apply. =
On which OS are you planning to use the agent?
1. linux
2. windows
3. darwin
default choice: [1]:
1
Trying to fetch the default region based on ec2 metadata...
If aws retry client will retry 1 timesAre you using EC2 or On-Premises hosts?
1. EC2
2. On-Premises
default choice: [1]:
1
Which user are you planning to run the agent?
1. cwagent
2. root
3. others
default choice: [1]:
1
Do you want to turn on StatsD daemon?
1. yes
2. no
default choice: [1]:
2
Do you want to monitor metrics from CollectD? WARNING: CollectD must be installed or the Agent will fail to start
```

```
Gr      "measurement": {
        "io_time"
      },
      "metrics_collection_interval": 60,
      "resources": [
        "*"
      ]
    },
    "mem": {
      "measurement": [
        "mem_used_percent"
      ],
      "metrics_collection_interval": 60
    },
    "swap": {
      "measurement": [
        "swap_used_percent"
      ],
      "metrics_collection_interval": 60
    }
  }
}

Please check the above content of the config.
The config file is also located at /opt/aws/amazon-cloudwatch-agent/bin/config.json.
Edit it manually if needed.
Do you want to store the config in the SSM parameter store?
1. yes
2. no
default choice: [1]:
2
Program exits now.
[ec2-user@ip-172-31-34-156 ~]$
```

i-0ae93952e8663f55e (CloudWatch-demo-22bbs0155)
PublicIP: 3.110.197.222 PrivateIP: 172.31.34.156

Start the CloudWatch agent:

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
```

```
-a fetch-config \
```

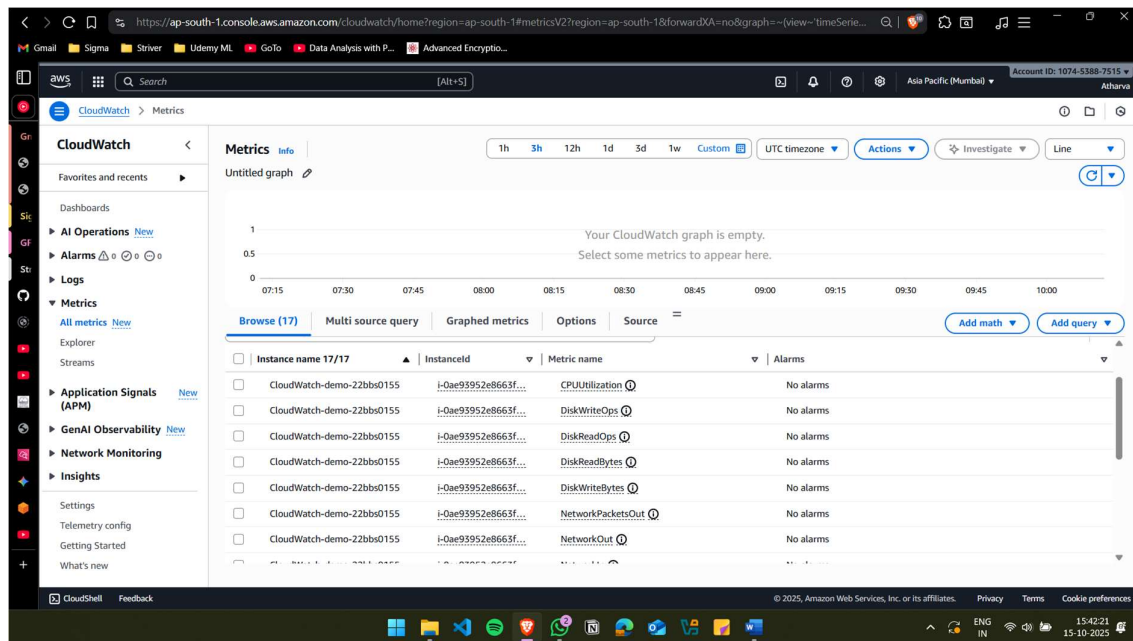
```
-m ec2 \
```

```
-c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json -s
```

```
Gr      Please check the above content of the config.
      The config file is also located at /opt/aws/amazon-cloudwatch-agent/bin/config.json.
      Edit it manually if needed.
      Do you want to store the config in the SSM parameter store?
      1. yes
      2. no
      default choice: [1]:
      2
      Program exits now.
      [ec2-user@ip-172-31-34-156 ~]$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
      -a fetch-config \
      -m ec2 \
      -c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json -s
      ***** processing amazon-cloudwatch-agent *****
      I [INFO] Trying to detect region from ec2 DI [EC2] Found active network interface I! imds retry client will retry 1 times
      Successfully fetched the config and saved in /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.d/file_config.json.tmp
      Start configuration validation...
      2025/10/15 10:01:38 Reading json config file path: /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.d/file_config.json.tmp ...
      2025/10/15 10:01:38 I! Valid json input schema.
      2025/10/15 10:01:38 DI ec2tagger processor required because append dimensions is set
      2025/10/15 10:01:38 DI delta processor required because metrics with diskio or net are set
      2025/10/15 10:01:38 DI ec2tagger processor required because append dimensions is set
      2025/10/15 10:01:38 Configuration validation first phase succeeded
      I! Detecting run as user...
      I! Trying to detect region from ec2
      DI [EC2] Found active network interface
      I! imds retry client will retry 1 times
      /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -schematest -config /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.toml
      Configuration validation succeeded
      amazon-cloudwatch-agent has already been stopped
      Created symlink /etc/systemd/system/multi-user.target.wants/amazon-cloudwatch-agent.service → /etc/systemd/system/amazon-cloudwatch-agent.service.
      [ec2-user@ip-172-31-34-156 ~]$
```

i-0ae93952e8663f55e (CloudWatch-demo-22bbs0155)
PublicIP: 3.110.197.222 PrivateIP: 172.31.34.156

Go to CloudWatch Console → Metrics → EC2 and verify data collection.



Create a CloudWatch Alarm:

Metric: CPU-Utilization

Condition: >70% for 2 consecutive periods

Notification: via SNS topic (send email alert).

The screenshot shows the 'Specify metric and conditions' step of the 'Create alarm' wizard. The left sidebar shows the progress: Step 1 (Specify metric and conditions), Step 2 (Configure actions), Step 3 (Add alarm details), Step 4 (Preview and create). The main area has a 'Metric' section with a 'Graph' preview and a 'Select metric' button. The 'Graph' section shows a preview of the metric or metric expression and the alarm threshold.

Specify metric and conditions

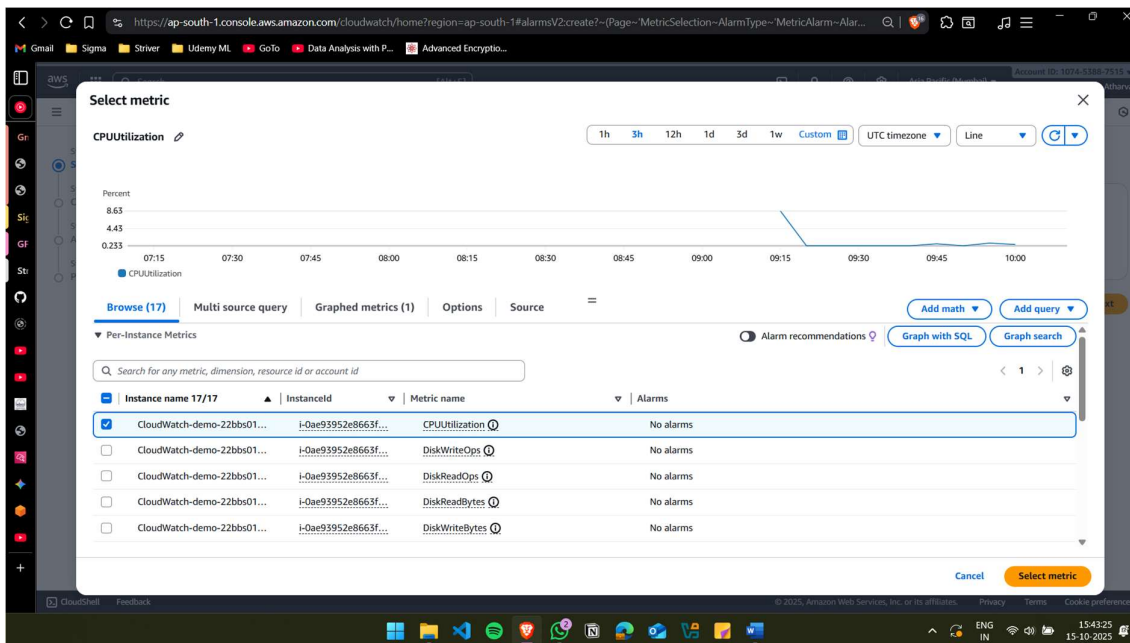
Metric

Graph

Preview of the metric or metric expression and the alarm threshold.

Select metric

Cancel Next



https://ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#alarmsV2:create?-(Page~MetricSelection~AlarmType~MetricAlarm~Alar...

CloudWatch Alarms Create alarm

Step 2: Configure actions

Step 3: Add alarm details

Step 4: Preview and create

Metric

Graph

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

Percent

70

35.1

0.233

07:15 07:30 07:45 08:00 08:15 08:30 08:45 09:00 09:15 09:30 09:45 10:00

CPUUtilization

Namespace: AWS/EC2

Metric name: CPUUtilization

Instanceld: i-0ae93952e8663f55e

Instance name: CloudWatch-demo-22bbs0155

Statistic: Average

Period: 5 minutes

Conditions

Threshold type: Static

Whenever CPUUtilization is...

Greater

Greater/Equal

Lower/Equal

Lower

than...

70

Additional configuration

Datapoints to alarm

Define the number of datapoints within the evaluation period that must be breaching to cause the alarm to go to ALARM state.

2 out of 2

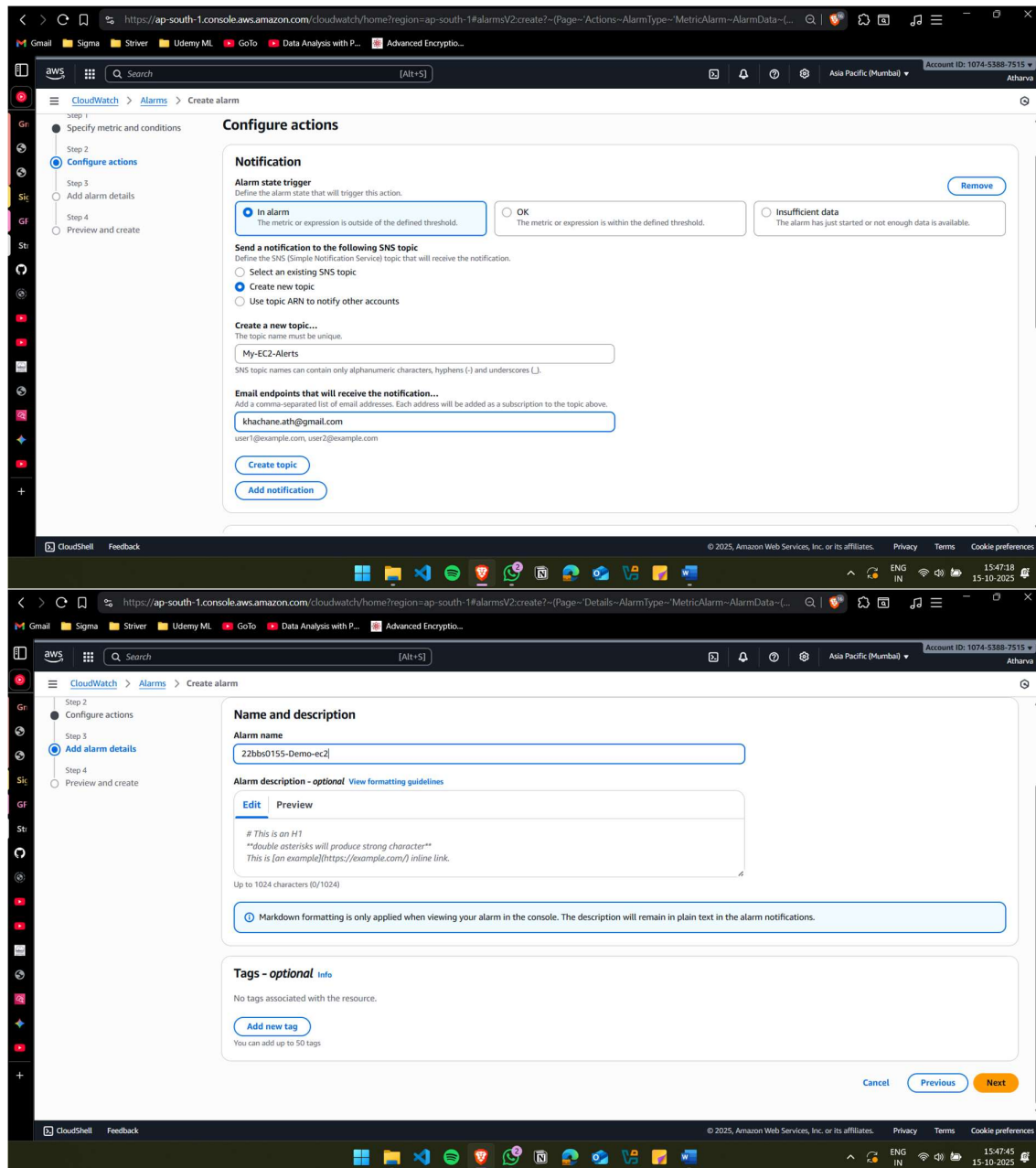
Missing data treatment

How to treat missing data when evaluating the alarm.

Treat missing data as missing

Cancel Next

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